Singularities: The Domain of Quantum Gravity

The beginning of a singularity is a place where the warping of space and time grows without bound. Where space warps and time warps become infinitely strong.

If we think of our universe's warped space as like the undulating surface of the ocean, then the beginning of a singularity is like the tip of a wave that is about to break, and the interior of the singularity is like the froth after it breaks (Figure 26.4). The smooth wave, before it breaks, is governed by smooth laws of physics, analogs of Einstein's relativistic laws. The froth after it breaks requires laws capable of dealing with frothing water, analogs of the laws of quantum gravity with their quantum foam.



Fig. 26.4. A singularity at the tip of an ocean wave that is about to break.

Singularities inhabit the cores of black holes. Einstein's relativistic laws predict them unequivocally, even though those laws can't tell us